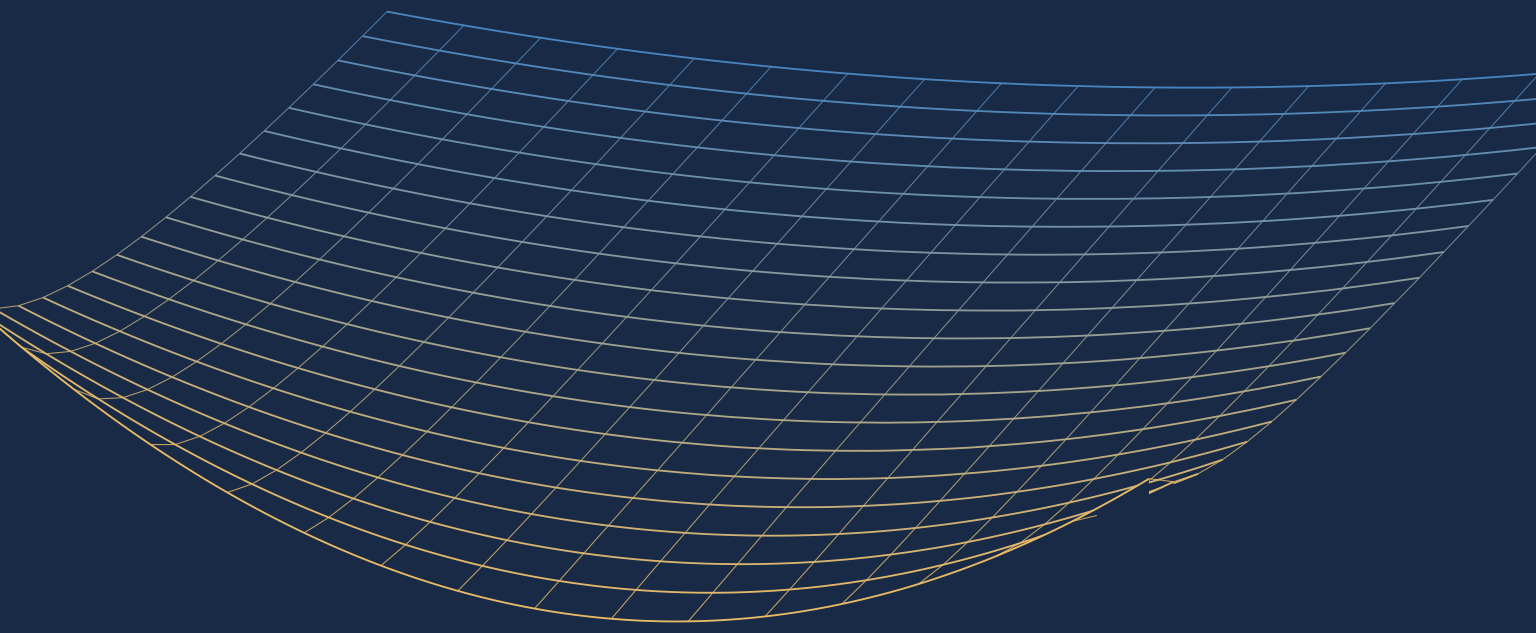
DRAFT EDITION

DERIVATIVES

CONTRACTS, MODELS, AND MARKETS

WULIN SUO



About the Cover

The cover is an equity implied volatility surface, the object of Chapter 19 and the one picture that says what this book is about.

The curves running across the sheet are smiles, one to a maturity: implied volatility against moneyness, out-of-the-money puts to the left, calls to the right. The lines running the other way join them along expiration, so the surface reads as a mesh rather than as a set of separate curves. Maturity runs back to front, two years at the back and six weeks at the front, and read in that direction you watch the skew steepen as expiration approaches: the most durable empirical regularity in equity option markets, and the one thing a single Black–Scholes volatility cannot accommodate, since it would draw this picture as a flat plane. The colors run with maturity, the amber of the near smiles cooling to blue at the long end; they are the two colors the interior uses for its worked examples and its case studies.

The shape is not artistic license. It is drawn from

$$\sigma(k,T) \;=\; 0.18 \;+\; \frac{0.11\,k^2}{T+0.30} \;-\; \frac{0.075\,k}{T+0.50},$$

with k the log-moneyness and T the maturity in years: a convexity term and a skew term, each decaying with maturity at its own rate. Curvature and tilt therefore fade at different speeds, which is what makes the surface twist as it flattens rather than simply subside.

That it is computed rather than drawn is the point, and it is the rule the whole book follows. Every plot here is generated in TEX from the formula the text derives, so no curve can drift out of step with the equation beside it; the cover is held to the same standard as the figures inside it.

The surface is also the natural emblem for the title. It is a market observable, quoted and traded in its own right. It is the output of a model, and the place where models are most visibly asked to do more than they can. And its shape is a fact about market structure: who buys crash protection, and who is left short it. Contracts, models, and markets meet here, which is why this is the picture on the front.

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Derivatives: Contracts, Models, and Markets

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Draft — November 13, 2026

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Derivatives: Contracts, Models, and Markets
Wulin Suo

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Typeset by the author in LaTeX. Every plot is drawn in TeX from the formulas the text derives, and no artwork is imported; the cover is described on page 2.

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Preface

This book teaches derivatives the way the subject is actually organized: instruments and the infrastructure that clears them, then options and their valuation, then volatility, credit, rates, and the places where models are asked to do more than they can. That sequence is not mine. It was established by John Hull in *Options, Futures, and Other Derivatives*, refined across eleven editions, and it is the order in which the subject is taught almost everywhere for the good reason that it works. A student who understands forward pricing before futures, futures before swaps, and binomial trees before Black–Scholes is a student who has never had to accept a result without proof. I have followed that sequence deliberately and I want to say so at the outset, because a reader who knows Hull's book will recognize the architecture and should not have to wonder whether the resemblance is accidental.

What follows is not a revision of that book, and it is not authorized by or affiliated with it. It is a new one, written because the subject moved. Professor Hull passed away on January 31, 2026, and the eleventh edition, dated 2022, is the last. The years around it were not quiet ones. LIBOR was replaced by overnight rates compounded in arrears, and with it the single discount curve that a generation of textbooks assumed. Collateral became the thing that determines what a derivative is worth, so that valuation adjustments moved from a specialist annex into the price itself. Volatility modeling absorbed a genuinely new idea in rough volatility, which changes how the path of volatility itself is modeled. Clearing houses became systemically central, and their margin models became a market risk in their own right. Commodities and crypto each acquired enough substance to deserve a chapter rather than a remark, and machine learning enough to deserve the online technical note that accompanies the book. *A book that stopped in 2022 would now mislead a practitioner about the mechanics of the instruments they trade every day*, and that is the gap this one is written to fill.

I owe Professor Hull a great deal more than a sequence, and I came to the subject through him. My first doctorate was in applied mathematics (stochastic control theory), and I spent the year after it as a post-doctoral fellow in the operations management group at the Faculty of Management, University of Toronto, now the Joseph L. Rotman School of Management. By the end of that year I had decided against a career in mathematics, without much idea of what else the training was for. Someone there told me that a professor named John Hull was well known in financial derivatives, and that a mathematical background was useful there. I bought his book, then in its second edition, and the subject turned out to be a great deal more interesting than a way of salvaging a training. He took me on as a doctoral student in finance at the University of Toronto. That copy was the second edition; this book is written after the eleventh.

Much of what I understand about how to explain a model, and about the difference between a model that fits and a model that is right, I learned from him. Readers who want the original statements of results this book uses will find his research papers in Appendix D, cited where the results are used, alongside the rest of the literature.

There is something further that no citation captures. More than any other book, his brought people into this subject. A great many of the practitioners and academics now working in derivatives first understood what a derivative was from its pages, and would say so without being asked; I am one of them. That is a rarer achievement than any particular result, and it is

the standard this book is written against rather than one it expects to meet.

Where this book departs from its predecessor, it does so on purpose, and the departures are the reason it exists.

It argues. A textbook can present a model and leave the reader to infer what it is worth. This one states what each model captures, where it fails, and how a practitioner would tell, often with the failure quantified. Chapter 21 measures how badly Heston misses the short-dated skew; Chapter 26 shows that a senior tranche's expected loss moves by a factor of twenty-eight for a tripling of the default rate. The claims are meant to be checkable, and where I make one I have tried to give the number that supports it.

Every plot is computed, not drawn. The plots are generated from the same formulas the text derives, so a curve cannot drift out of step with the equation beside it. Where a worked example runs long, the script that produced it is kept, so the figure can always be recomputed.

It is bilingual, and the Chinese edition is a superset rather than a translation. It covers everything here, on the same terms, and adds the domestic market where it differs (China's benchmark architecture, its bond futures conventions, its structured products), because a Chinese reader needs both the international convention and the local one, and needs to understand the reason for the gap.

And it takes market structure seriously as subject matter. Clearing, margin, collateral, auctions, and the definitions inside contracts are not preliminaries to the mathematics. They are where a large share of actual losses come from, and several chapters are built around episodes (2008, the 2020 dash for cash, Archegos, the LME nickel squeeze, Credit Suisse's additional tier 1) in which the mechanism mattered more than the model.

The mathematical level is deliberately unchanged from its predecessor's: accessible, and worked through with intuition and numbers rather than asserted.

Five practical notes follow on how the book is put together.

Elective material. A dagger on a chapter or section title (Chapters 14, 21, 28, 30 and 32, and sections of Chapters 20, 38 and 39) marks material aimed at readers with stronger quantitative preparation. The mark appears in the table of contents and in the running head as well as on the heading, so a path through the book can be planned from the contents alone. Elective material can be skipped without breaking continuity: an MBA reader can pass from Chapter 13 to Chapter 15 without Chapter 14, for instance. A financial-engineering reader should not skip it, and nothing in it is optional in the sense of being less settled or less used—it is harder, not softer. Section 1.13 maps the dependencies between chapters and is the place to start if you are assembling a path rather than reading straight through.

Technical notes. Derivations that would interrupt the main argument are collected in a companion set of technical notes, published separately and referenced in the text as [TN15], numbered to match the chapter that defers to them; TN41, on machine learning, stands alone and follows the last chapter. The main text states results and explains why they hold; the notes prove them. The two the argument leans on hardest are reproduced in Appendix A, so a reader working from the book alone is never stopped by one.

Examples and case studies. Every formula whose magnitude matters is carried through actual numbers, because a reader who has not seen a convexity adjustment expressed in basis points does not yet know whether it is worth computing. Several worked examples deliberately verify a result two ways, and those cross-checks are part of the method being taught rather than decoration. The case studies carry the episodes: Metallgesellschaft, Amaranth, Archegos, the LME nickel squeeze, and the rest.

Companion notebooks. Twenty-nine chapters (3, 9 and 13–39) come with a Python notebook that reproduces their numbers; the online note on machine learning, TN41, has one as well. The code is written out rather than delegated: a calculation usually left to a library is

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set down step by step, because seeing it is the point of reading it. No chapter depends on its notebook, since each is complete as it stands, but a reader who wants to change a parameter and watch what moves will get further with one open. The two editions run identical code; only the narrative and the printed labels are translated.

Boxed equations. A box around an equation marks a result that later chapters build on and that a reader should be able to reproduce from memory. A chapter has one or two at most, and many have none; intermediate steps, approximations and special cases are never boxed, however often they are used.

Errors that remain are mine. I would be grateful to hear of them.

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† indicates an elective chapter or section; the preface explains what that means.

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Notation

The literatures this book spans grew up separately and their symbols collide; these are the conventions used here, followed by the symbols that carry more than one meaning.

PRICES, RATES, AND TIME

t	the present, or a general date
T	maturity, expiration, or settlement date
τ	accrual fraction between two dates, in years
S, S_0, S_T	spot price of the underlying: generally, now, at maturity
F, F_0	forward or futures price; a forward rate in the rates chapters
K	strike price, or the delivery price of a forward
r	continuously compounded risk-free rate
q	dividend yield
r_f	foreign risk-free rate, the yield on a currency, so $q = r_f$
y	convenience yield on a commodity
L, N	notional principal
W, dW	Wiener process and its increment

OPTIONS AND VOLATILITY

c, p	prices of a European call and put (Chapters 11, 15, 16)
C, P	prices of an American call and put (Chapter 11); P standing alone is never a bond price
σ	volatility; implied volatility where the context is a quote
$N(x)$	standard normal cumulative distribution function
$N'(x)$	standard normal density, $e^{-x^2/2}/\sqrt{2\pi}$; written φ in Chapters 29 and 36, ϕ in Chapters 24 and 25
d_1, d_2	the arguments of the Black–Scholes–Merton and Black formulas
Δ, Γ, Θ, $\mathcal{V}$	delta, gamma, theta, vega
H	barrier level
κ	speed of mean reversion of variance in a stochastic volatility model
H (Chs. 14, 21)	Hurst parameter of fractional Brownian motion, and of a rough volatility model

TERM STRUCTURE

$P(t,T)$	price at t of a zero-coupon bond maturing at T
DF_i	discount factor to date t_i (Chapters 4, 7 and 9)
$f(t,T)$	instantaneous forward rate (Chapters 31 and 32)
$A(t,T)$, $B(t,T)$	the coefficients of an affine bond price, $P = A\,e^{-Br}$
$B(t)$	value at t of one unit invested from time 0 at the overnight rate used for discounting; the accumulation account
A	annuity, or PV01 leg, of a swap
a	speed of mean reversion of the short rate
Q	the risk-neutral measure
Q^T, Q^N	the T-forward measure and the measure induced by numeraire N
E, $\mathbb{E}$	expectation, with the measure as a superscript (E^Q, E^T); the two forms are interchangeable
Λ	a Radon–Nikodym derivative: dQ/dP in Chapter 28, and the ratio of two forward measures in Chapter 30; Λ_t is its restriction to $\mathcal{F}_t$, the density process

CREDIT

λ	hazard rate, or default intensity
τ	time of default
$Q(t)$	survival probability to t (Chapters 24–26); not the measure
R	recovery rate
A	risky annuity: the present value of one basis point paid until default
ρ	default correlation

RISK

c	confidence level of VaR and ES; written q in the WCDR of Chapters 24 and 26
μ	drift, or mean
ρ	correlation
β	sensitivity to a common factor

SYMBOLS WITH MORE THAN ONE MEANING

- ρ: a *correlation* nearly everywhere (including price against variance in a stochastic volatility model), and the Greek *rho* in Chapter 18.
- H: a *CSA threshold* in Chapter 9, a *barrier level* in Chapters 34 and 35, the *Hurst parameter* in Chapters 14 and 21, a *futures price* in Chapter 28, a *haircut* in Chapter 27; no chapter uses two at once.
- λ: a *hazard rate* in Chapters 9, 24 and 25, the *market price of risk* in Chapter 28, a *jump intensity* in Chapter 21, the *EWMA decay* in Chapter 38, the *barrier exponent* in Chapter 34.
- B: the *accumulation account* $B(t)$ (Chapters 28 and 30) and an *affine coefficient* $B(t,T)$ (Chapter 31); the arguments separate them.
- τ: an *accrual fraction* in the rates chapters, the *time of default* in Chapters 9, 24 and 25, the *first call date* of an autocallable in Chapter 35.
- A: an *annuity* in the swap chapters, a *risky annuity* in the credit chapters; the same construction, stopping or not at default.
- φ, ϕ: the normal density in Chapters 24, 25, 29 and 36, Heston's *characteristic function* in Chapter 21, the call/put *sign* ± 1 in Chapter 34, a deterministic *shift* $\phi(t,T)$ in Chapter 32; Φ_k in Chapter 32 is a volatility *multiplier*, not a distribution function.

The normal distribution function is $N(\cdot)$, not $\Phi(\cdot)$, except in Chapter 36's weather options. $P(t,T)$ is always a bond price; P alone is the American put of Chapter 11, or in Chapters 19, 21 and 22 a European put as a function of strike, $P(K)$.

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